

Electric Flux

Prepared by:

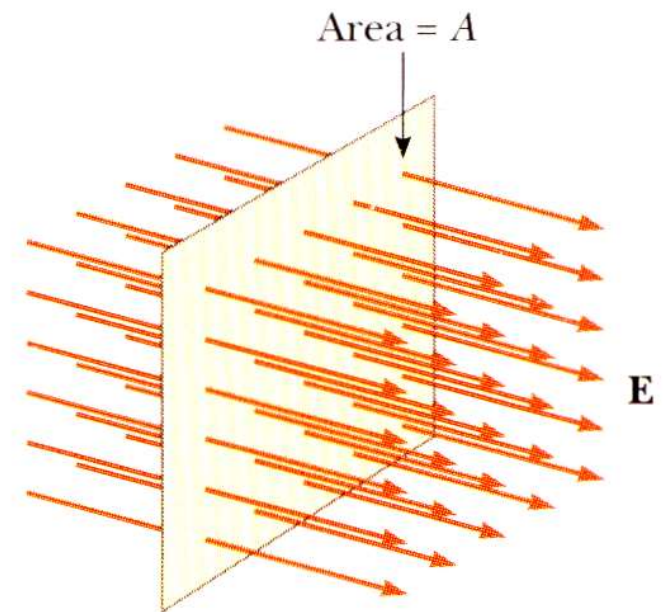
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Electric Flux

- Electric Flux of a certain surface is the number of the electric field line outward from the surface
- Scalar quantity, φ
- It will be positive if the electric field lines are outward from the surface ($\varphi = +ve$)
- It will be negative if the electric field lines are inward to the surface ($\varphi = -ve$)

Electric Flux

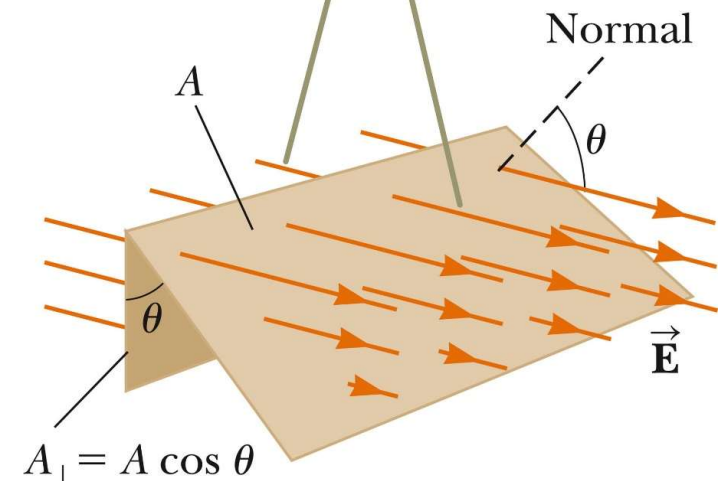
- $|\underline{E}|$ is the number of electric field lines per unit normal Area
- Then, $\varphi = |\underline{E}|A_N$
- where A_N is the normal area that is perpendicular to the field.
- Units of φ is Nm^2/C



Electric Flux

- If $\underline{E}$ is not perpendicular to the surface
- $\varphi = |\underline{E}|A_N = |\underline{E}|A \cos(\theta)$
- $\varphi = \underline{E} \cdot \underline{A}$
- $\underline{A}$ is the area vector that its magnitude is the surface area and its direction is perpendicular over it outwards

The number of field lines that go through the area $A_{\perp}$ is the same as the number that go through area A .



Electric Flux

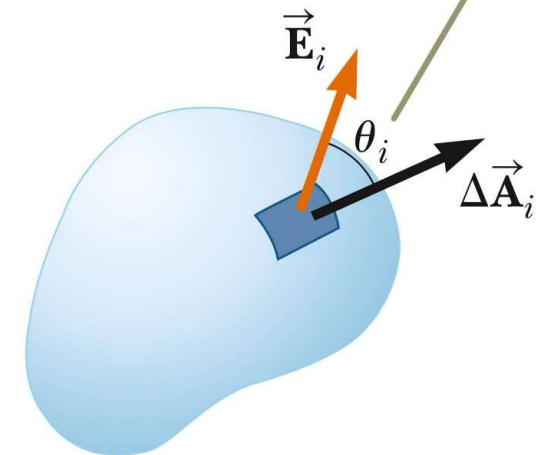
- The flux is a maximum when the surface is perpendicular to the field.
 - $\theta = 0^\circ$
- The flux is zero when the surface is parallel to the field.
 - $\theta = 90^\circ$
- If the field varies over the surface, $\Phi = EA \cos \theta$ is valid for only a small element of the area.

Electric Flux

$$\therefore \Phi_E = \lim_{\Delta A_i \rightarrow 0} \sum_i \mathbf{E}_i \cdot \Delta \mathbf{A}_i$$

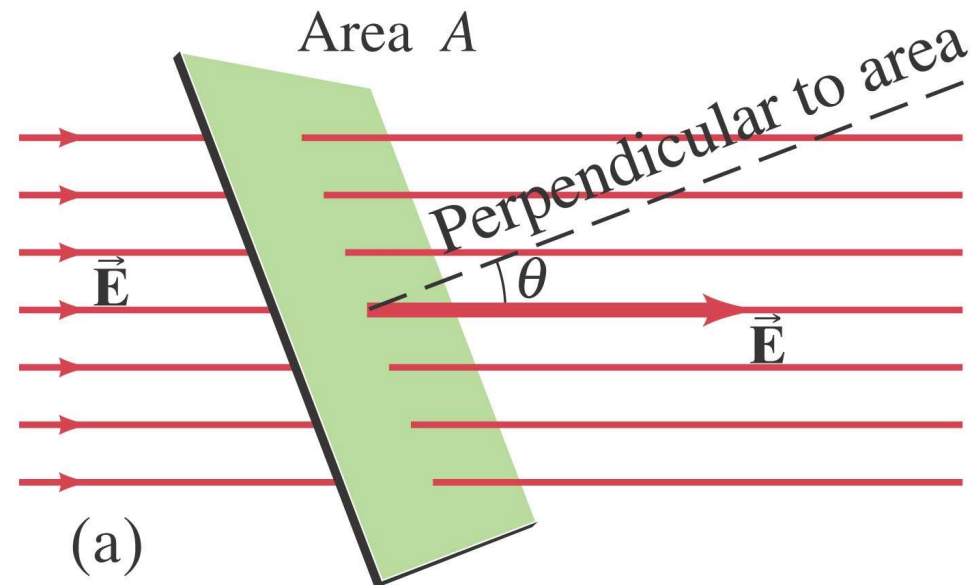
$$\therefore \Phi_E = \int \mathbf{E} \cdot d\mathbf{A}$$

The electric field makes an angle θ_i with the vector $\Delta \vec{\mathbf{A}}_i$, defined as being normal to the surface element.



Example 1:

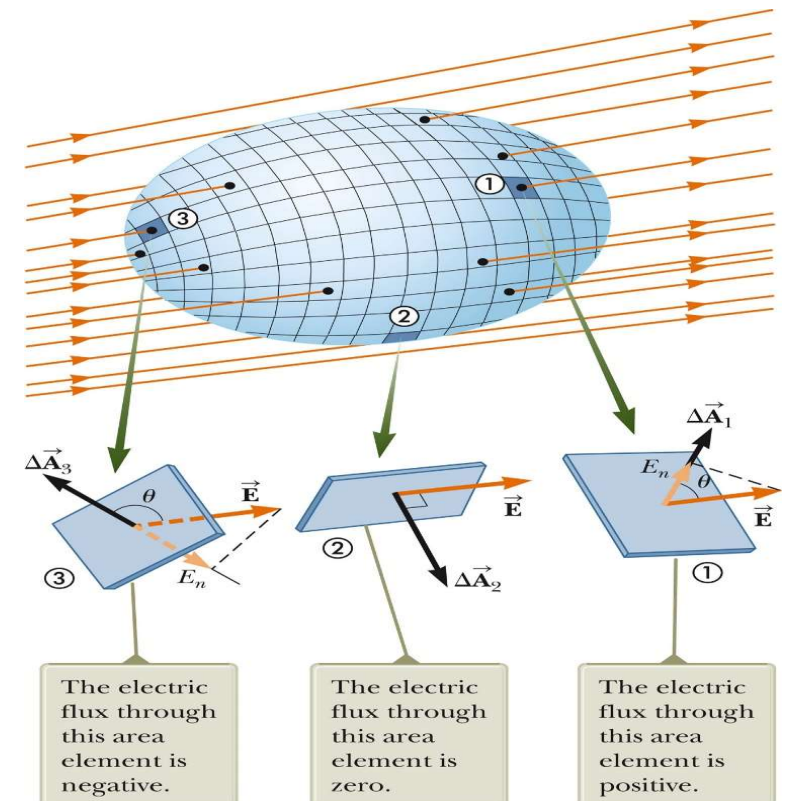
- Calculate the electric flux through the rectangle shown. The rectangle is 10 cm by 20 cm, the electric field is uniform at 200 N/C, and the angle θ is 30° .



The flux is $EA \cos \theta = 3.5$
 $\text{N}\cdot\text{m}^2/\text{C}.$

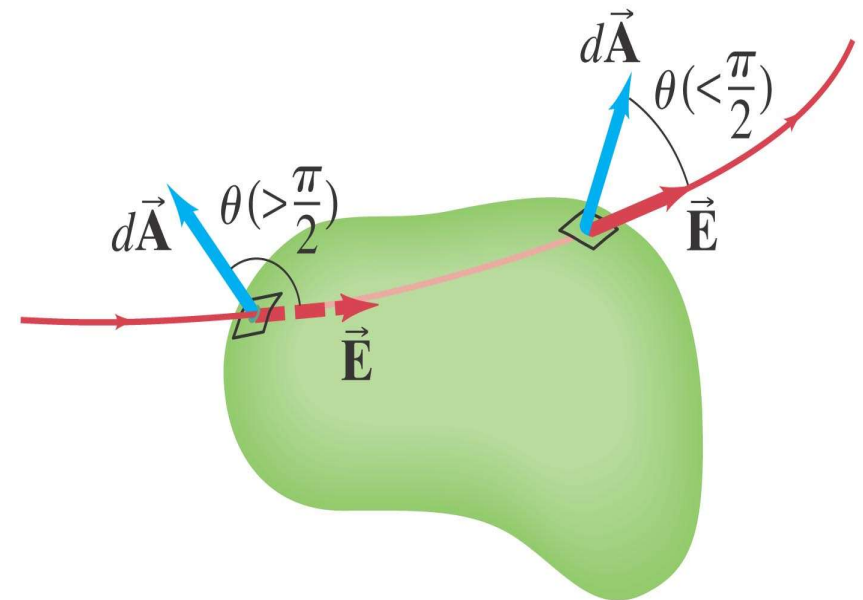
Electric Flux, Closed Surface

- At (1), the field lines are crossing the surface from the inside to the outside; $\theta < 90^\circ$, Φ is positive.
- At (2), the field lines graze surface; $\theta = 90^\circ$, $\Phi = 0$
- At (3), the field lines are crossing the surface from the outside to the inside; $180^\circ > \theta > 90^\circ$, Φ is negative.



Electric Flux, Closed Surface

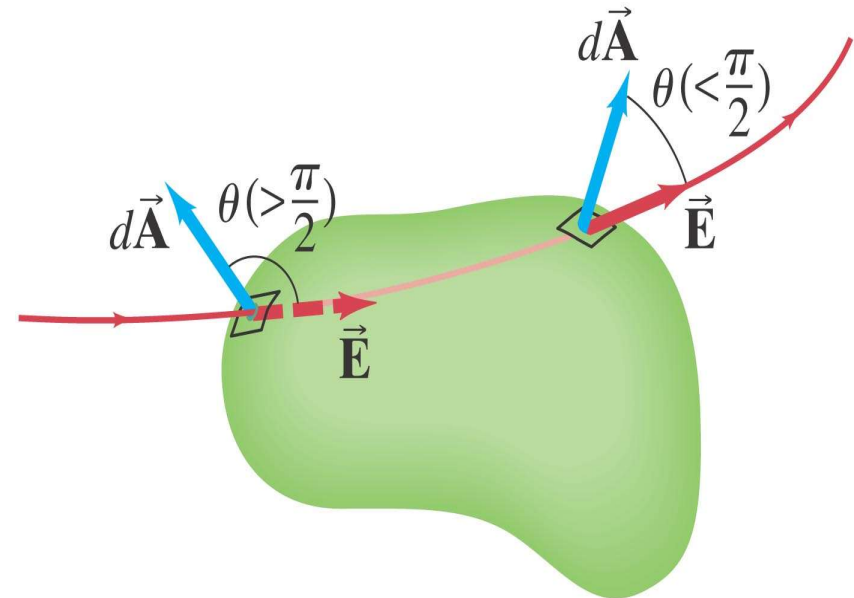
- The **net** flux through the surface is proportional to the net number of lines leaving the surface.
 - This net number of lines is the number of lines leaving the surface minus the number entering the surface.
 - This is proportional to the total charge by enclosed the surface



Electric Flux, Closed Surface

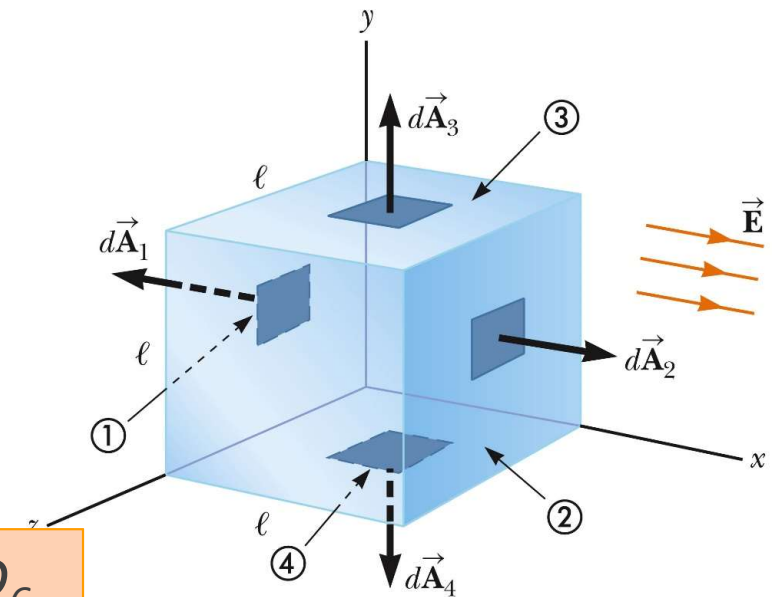
- Gauss' law

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0},$$



Example 2:

- Consider a uniform electric field $\vec{E}$ that is oriented in the x direction in empty space. A cube of edge length ℓ , is placed in the field, oriented as shown. Find the net electric flux through the surface of the cube

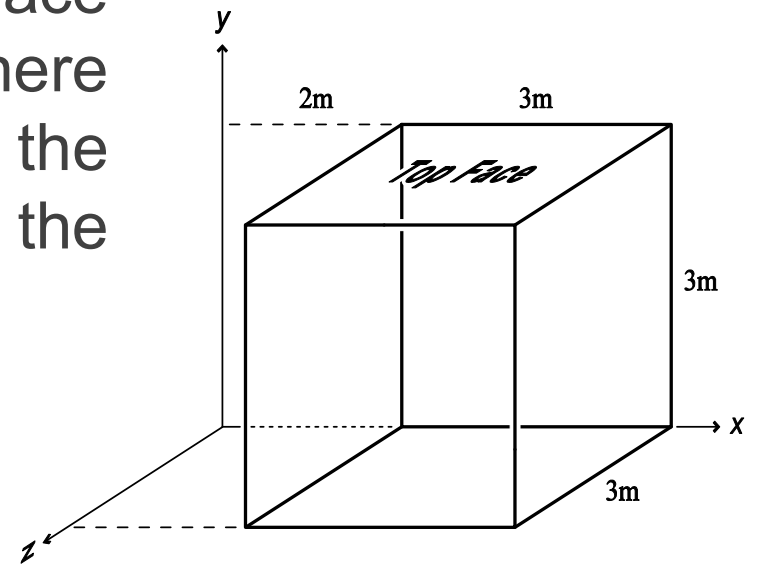


$$\Phi_1 = -E\ell^2, \Phi_2 = E\ell^2, \Phi_3 = \Phi_4 = \Phi_5 = \Phi_6 = 0$$

Therefore, $\Phi_{total} = 0$

Example 3:

The electric field in the region of space shown is given by $E = (8\mathbf{i} + 6\mathbf{j})$ N/C where y is in m. What is the magnitude of the electric flux through the Top face of the cube shown?



$$\text{Solution: } \Phi = E \cdot A_{\text{top}} = (8\mathbf{i} + 6\mathbf{j}) \cdot (3 \times 3 \mathbf{j}) = 54 \text{ Nm}^2/\text{C}$$

Check that

<https://www.youtube.com/watch?v=AXoleCOvjrE>